



Chapter 1 — Executive Summary & Vision

1.1 What is DOIP?

The **Defense Operations Intelligence Platform (DOIP)** is a browser-based command and decision-support application designed for **military simulation, training, and defense operations planning**. It gives a single operator (or a small team) a unified, real-time operational picture composed of:

- An **interactive live map** showing simulated friendly units, drones, supply depots, incidents, and risk overlays
- A **real-time event feed** of incoming reports, sensor readings, and alerts
- **AI-generated situation summaries**, mission briefs, and recommendations
- **Logistics, personnel-health, and weather dashboards** driven by synthetic data
- **Mission planning tools** with route optimization and constraint handling
- **Predictive analytics** that forecast resource consumption and mission outcomes

Every data point in the system — every GPS coordinate, sensor reading, weather change, and incident — is produced by a built-in **simulation engine**. There is no hardware, no IoT, no drones, and no connection to any real-world system. DOIP is pure software, end to end.

1.2 The Problem DOIP Addresses

Command-and-control style software sits at the intersection of several hard engineering problems that are rarely solved together in one system:

1. **Information fragmentation.** In real military and training operations, critical information lives in separate tools — maps in one app, reports in chat, logistics in spreadsheets, weather in a browser tab. Operators waste cognitive effort *assembling* the picture rather than *acting* on it.

2. **Data overload without synthesis.** Streams of raw reports ("Team Bravo delayed", "vehicle movement observed", "medical evacuation requested") arrive faster than a human can prioritize them. Raw data $\neq$ situational awareness.
3. **Planning without feedback.** Route and mission planning is usually done on static maps that don't account for live conditions like weather, incidents, or restricted zones.
4. **No safe training environment.** Training operators on real systems is expensive, risky, and often impossible. A high-fidelity *simulated* environment lets people practice decision-making with zero real-world consequences.
5. **Reactive rather than predictive operations.** Most operations tooling tells you what *has happened*. Very little tells you what is *about to happen* — supply shortages, mission delays, maintenance failures.

1.3 Why This Project Is Needed

As a product concept

- **Defense forces** need exactly this class of tool: a common operational picture + AI triage + logistics forecasting for border surveillance, convoy movement, casualty evacuation, and force sustainment. DOIP's simulation-only design makes it a safe, legal training and planning analogue of real C2 systems.
- **Military training organizations** need repeatable, scriptable exercises. DOIP's scenario engine can replay the same operation a hundred times for a hundred trainees — with an adaptive red-team opponent (Ch. 16).
- **HADR remains in scope as a military mission.** Disaster-relief deployments (aid to civil authority) are a standing task of armed forces worldwide and the only scenario class with open real-world outcome data for benchmarking (Ch. 14).
- **Planning teams** need what-if analysis: "If the storm hits Sector C at 14:00, which routes survive and which supplies run out?"

As an engineering demonstration

DOIP is deliberately designed to exercise nearly every advanced discipline in modern software engineering within a single coherent product:

Discipline	Where it appears in DOIP
Real-time systems	WebSocket push, live map updates, event streaming
Distributed architecture	Service decomposition, message brokers, API gateway
GIS / geospatial engineering	PostGIS, MapLibre, spatial queries, route computation
Applied AI — LLMs	Report summarization, operator Q&A, mission briefs
Applied AI — RAG	SOP knowledge base with cited retrieval
Applied AI — Computer Vision	YOLO object detection on prerecorded aerial footage
Machine learning	Forecasting with XGBoost / Prophet / LSTM
Algorithms	A*, Dijkstra, risk scoring, constraint-based planning
Data engineering	Event pipelines, time-series storage, replay
DevOps	Docker, Kubernetes, CI/CD, Prometheus, Grafana
Security engineering	JWT, RBAC, audit logging, rate limiting

Very few portfolio projects touch more than two or three of these. DOIP touches all of them **with a visible, demoable payoff for each**.

1.4 Vision Statement

One screen, total awareness. DOIP turns a flood of simulated operational data into a clear, live, decision-ready picture — and pairs the human operator with an AI assistant that reads everything, forgets nothing, and always shows its sources.

Think of DOIP as combining the DNA of:

- **Google Maps** — fluid, interactive geospatial visualization
- **Grafana** — dense, real-time operational dashboards
- **ChatGPT** — conversational AI that explains and summarizes
- **Flightradar24** — live tracking of moving entities
- **PagerDuty / incident management** — structured alerting and response workflows

...fused into a single integrated application.

1.5 Target Users & Personas

Persona	Role	Primary needs
The Operator	Watches the live dashboard during a simulation run	Fast comprehension, alert triage, AI summaries
The Mission Planner	Designs missions and routes before/during a run	Drawing tools, route optimization, constraint checks
The Logistics Analyst	Monitors supplies and predicts shortages	Resource dashboards, forecasts, threshold alerts
The Exercise Director (Admin)	Authors scenarios, starts/stops simulations, reviews performance	Scenario editor, replay, after-action analytics, user management
The Trainee	Learns operational decision-making in a safe environment	Guided scenarios, AI explanations, performance feedback

1.6 Core Use Cases

1. **Live monitoring** — Operator starts a scenario; units appear and move on the map; incidents fire; the operator watches, filters, and drills into events.
2. **AI situation reporting** — Every N minutes (or on demand), the AI condenses all recent events into a short prioritized situation report.
3. **Mission planning** — Planner draws a search zone, sets waypoints, and requests an optimized route that avoids restricted areas and weather-degraded terrain.
4. **Threat & anomaly detection** — The vision service processes prerecorded aerial/disaster footage, detects objects (vehicles, people, smoke, fire, damage), scores risk, and plots detections on the map.
5. **Logistics forecasting** — The analytics service predicts fuel/medical/food shortfalls hours before they occur and raises early warnings.
6. **Knowledge assistance (RAG)** — Operator asks “What is the SOP for medical evacuation in low visibility?” and receives a cited answer drawn from the loaded document base.
7. **After-action review** — When the simulation ends, the platform generates a performance report: response times, resource efficiency, decision timeline, and AI-written narrative summary.

1.7 Goals & Objectives

Primary goals

- G1: Deliver a working, demoable core platform (map + simulation + events + dashboard) early.
- G2: Integrate a genuinely useful AI layer (summarization + RAG assistant) grounded in platform data.
- G3: Demonstrate real-time architecture: sub-second event propagation from simulator to browser.
- G4: Demonstrate geospatial depth: PostGIS-backed spatial queries and algorithmic route planning.
- G5: Ship with production-grade practices: containerization, CI/CD, monitoring, auth, and docs.

Secondary goals

- G6: Computer-vision pipeline on prerecorded footage with map-plotted detections.
- G7: Predictive analytics for logistics and mission outcomes.
- G8: Scenario authoring and replay for repeatable training exercises.

1.8 Explicit Non-Goals



These boundaries are permanent design constraints, not future roadmap items.

- **No hardware integration.** No IoT devices, drones, radios, GPS receivers, cameras, or wearables — ever. All sensor data is synthetic.
- **No real-world control.** DOIP issues no commands to anything physical. It is display, planning, and simulation only.
- **No weapons or targeting functionality.** Scenario content is modeled on defensive and support operations — patrol, surveillance, convoy escort, casualty evacuation, logistics, and military HADR deployments — never offensive strike planning.

- **No classified or real operational data.** All maps are OpenStreetMap; all documents in the RAG base are public or self-authored training material.
- **No multi-region production scale.** This is a single-deployment demonstration platform, not a hardened enterprise SaaS.

1.9 What Success Looks Like

- A visitor can watch a **3-minute live demo**: scenario starts → units move → storm rolls in → incident fires → AI produces a situation report → operator plans a route around the danger zone → logistics forecast updates.
- The codebase reads like a production system: typed, tested, containerized, documented, observable.
- Each of the 7 phases has a tagged release with its own demo recording and changelog.
- The architecture story is explainable in one sentence per layer — and every technology choice has a written justification (Chapter 4).